Stage 1 Report — ASD Hybrid 10-fold

STAGE 1 REPORT - workspace, environment and dataset audit

Project: Reliability-Weighted Multimodal AI for Early Autism Risk Screening in Children (BITE308L). Project dir: `/home/score/asd_hybrid_10fold`. Raw data (read-only): `/home/score/Documents/AI_project/dataset`. Seed: 42.

1. Summary

- Built the project skeleton, the results layout (`results/README.md`) and a verified CUDA environment (`.venv`).
- Audited all 6 raw datasets read-only (`scripts/01_audit_datasets.py`). Outputs are in `data_registry/` (json, md, contact sheets, licence text).
- Found serious data problems: D03 is a copy of D02; the D05 labels are a formula of A1-A10; the D05 Combined file has 728 wrong labels; 40% of D01 rows have no child ID and 10 D01 file pairs share rows.
- Recorded 12 dataset decisions with evidence (`data_registry/DATASET_DECISIONS.md`).
- pytest: 24 passed; raw data unchanged (snapshot START == END).

2. Environment

Source: `data_registry/environment.txt`.

item	value
GPU	NVIDIA RTX A2000 12GB
GPU memory	536MiB / 12282MiB (nvidia-smi used/total); torch total = 11.618 GiB
NVIDIA driver	580.178.04 (driver CUDA 13.0)
torch CUDA / cuDNN	13.0 / 92400

item	value
Python	3.12.3
torch / torchvision / timm	2.14.1+cu130 / 0.29.1+cu130 / 1.0.30
cuda.is_available()	True
4096x4096 fp32 matmul	29.83 ms (4.61 TFLOP/s)
venv	`/home/score/asd_hybrid_10fold/.venv` (`--system-site-packages`; numpy/scipy/matplotlib/pillow/pyyaml come from the system and are pinned via `logs/pip_constraints.txt`)
requirements	`requirements.txt` (exact versions of the 13 required packages)

3. Project skeleton (depth 2, excluding .venv and cache)

```
asd_hybrid_10fold/
├── .gitignore
├── configs/
│   └── .gitkeep
├── data_registry/
│   ├── .gitkeep
│   ├── D01_cv_feasibility.json
│   ├── D01_dedupe_key_check.json
│   ├── D01_eye_tracking_audit.json
│   ├── D01_eye_tracking_audit.md
│   ├── D01_eye_tracking_licence_citation.txt
│   ├── D02_facial_images_A_audit.json
│   ├── D02_facial_images_A_audit.md
│   ├── D02_facial_images_A_licence_citation.txt
│   ├── D02_facial_images_A_samples.png
│   ├── D03_facial_images_B_audit.json
│   ├── D03_facial_images_B_audit.md
│   ├── D03_facial_images_B_licence_citation.txt
```

```
│   ├── D03_facial_images_B_samples.png
│   ├── D04_autism_prediction_audit.json
│   ├── D04_autism_prediction_audit.md
│   ├── D04_autism_prediction_licence_citation.txt
│   ├── D05_toddler_screening_audit.json
│   ├── D05_toddler_screening_audit.md
│   ├── D05_toddler_screening_licence_citation.txt
│   ├── D06_early_autism_audit.json
│   ├── D06_early_autism_audit.md
│   ├── D06_early_autism_licence_citation.txt
│   ├── DATASET_DECISIONS.md
│   ├── DATASET_REGISTRY.md
│   ├── environment.txt
│   ├── raw_snapshot_end.tsv
│   └── raw_snapshot_start.tsv
├── logs/
│   ├── .gitkeep
│   ├── pip_constraints.txt
│   ├── pip_install.log
│   ├── stage01_audit.log
│   ├── stage01_facts.txt
│   ├── stage01_pytest.txt
│   └── stage01_snapshot_compare.txt
├── reports/
│   ├── .gitkeep
│   └── STAGE_01_REPORT.md
├── requirements.txt
├── results/
│   ├── .gitkeep
│   └── README.md
├── scripts/
│   ├── .gitkeep
│   ├── 00_snapshot_raw.py
│   ├── 01_audit_datasets.py
│   ├── 01_build_registry.py
│   ├── 01_check_cv_feasibility.py
│   ├── 01_check_dd02_key.py
│   ├── 01_check_environment.py
│   ├── 01_git_commit.sh
│   └── 01_write_report.py
├── splits/
│   └── .gitkeep
├── src/
│   ├── data/
│   ├── evaluation/
│   ├── models/
│   ├── training/
│   └── utils/
```

```
└── tests/
    ├── .gitkeep
    └── test_stage01.py
```

4. Dataset registry

Folder -> id mapping (all inside `/home/score/Documents/AI_project/dataset/`):

- **D01_eye_tracking** <- `Eye-Tracking Dataset`
- **D02_facial_images_A** <- `Autism Spectrum Disorder Detection using Facial Traits Dual Framework DenseRes with K-Fold Cross Validation (DFDK) for Enhanced Classification Accuracy`
- **D03_facial_images_B** <- `AutismDataset`
- **D04_autism_prediction** <- `Autism_prediction dataset`
- **D05_toddler_screening** <- `Autism_dataset`
- **D06_early_autism** <- `early_autism`

The folder truncated in the file manager is, in full: `Autism Spectrum Disorder Detection using Facial Traits Dual Framework DenseRes with K-Fold Cross Validation (DFDK) for Enhanced Classification Accuracy` (= D02).

id	folder name	modality	n samples	n subjects	classes and counts	duplicates found	leakage findings	suitable for
D01_eye_tracking	`Eye-Tracking Dataset`	eye-tracking time-series CSV (gaze)	2,252,943 gaze rows in 25 CSVs; 901,786 rows have no participant ID ({'Unidentified(Neg)': 878362, 'Unidentified(Pos)': 23424})	59 in metadata; 57 with gaze rows (no gaze: IDs [12, 16])	metadata {'TD': 30, 'ASD': 29}; with gaze {'ASD': 27, 'TD': 30}	329,034 rows repeat a (RecordingTime, TimeOfDay, Participant, Trial) key; 15 files share rows; 10 file pairs share DD-02-key rows. DD-02 numeric key, identified rows: 1,351,157 -> 1,103,660 (D01_dedupe_key_check.json)	CARS Score missing/n by class {'ASD': {'cars_missing': 0, 'n': 29}, 'TD': {'cars_missing': 30, 'n': 30}} -> CARS is a label proxy and must never be a feature. Participant IDs recur across files -> split by participant.	10 hybrid architectures (gaze-image sequences), participant-level folds

id	folder name	modality	n samples	n subjects	classes and counts	duplicates found	leakage findings	suitable for
D02_facial_images_A	Autism Spectrum Disorder Detection using Facial Traits Dual Framework DenseRes with K-Fold Cross Validation (DFDK) for Enhanced Classification Accuracy	facial images (RGB jpg)	2940 images (train: {'autistic': 1193, 'nonautistic': 1193}, valid: {'autistic': 127, 'nonautistic': 127}, test: {'nonautistic': 150, 'autistic': 150}); 0 unreadable; size 114-2471 x 126-2872 px	no subject IDs	{'autistic': 1470, 'nonautistic': 1470}	exact md5: 0 groups; near (phash<=4): 11 pairs, 5 across splits {'train/valid': 2, 'test/train': 3}, 0 cross-class. vs D03_facial_images_B: 2541 byte-identical, 2940/2940 D03 images have a phash<=4 match in D02 (same class: 2940)	n/a (images). No subject IDs -> same child may fall in train and test; near-duplicates across splits listed.	10 hybrid architectures
D03_facial_images_B	AutismDataset	facial images (RGB jpg)	2940 images (none: {'autistic': 1470, 'nonautistic': 1470}); 0 unreadable; size 114-2471 x 126-2872 px	no subject IDs	{'autistic': 1470, 'nonautistic': 1470}	exact md5: 0 groups; near (phash<=4): 11 pairs, 0 across splits , 0 cross-class. vs D02_facial_images_A: 2541 byte-identical, 2940/2940 D03 images have a phash<=4 match in D02 (same class: 2940)	n/a (images). No subject IDs -> same child may fall in train and test; near-duplicates across splits listed.	10 hybrid architectures in principle, BUT it is the same 2940 images as D02 -> use D02 only; never treat D03 as an independent dataset
D04_autism_prediction	Autism_prediction dataset	questionnaire / screening table	test.csv 200x21; train.csv 800x22	no subject IDs beyond per-row ID columns (test.csv: ID; train.csv: ID); cannot tell whether rows	test.csv: NO LABEL (unlabelled); train.csv: {'0': 639, '1': 161}	test.csv: 0 full / 0 excl. ID; train.csv: 0 full / 0 excl. ID	train.csv: No exact single-feature rule over the whole file; best is 'ASD if sum(A1_Score..A10_Score) >= 10' accuracy 0.8625 (majority baseline	tabular baselines only (LR, RF, GB, MLP) - NOT the CNN/ViT hybrids

id	folder name	modality	n samples	n subjects	classes and counts	duplicates found	leakage findings	suitable for
				are distinct people			0.79875)	
D05_toddler_screening	Autism_dataset	questionnaire / screening table	Autism_Screening_Data_Combined.csv 6075x15; Toddler Autism dataset July 2018.csv 1054x19	no subject IDs beyond per-row ID columns (Autism_Screening_Data_Combined.csv: none; Toddler Autism dataset July 2018.csv: Case_No); cannot tell whether rows are distinct people	Autism_Screening_Data_Combined.csv: {'NO': 4271, 'YES': 1804}; Toddler Autism dataset July 2018.csv: {'Yes': 728, 'No': 326}	Autism_Screening_Data_Combined.csv: 847 full / 847 excl. ID; Toddler Autism dataset July 2018.csv: 0 full / 79 excl. ID	Autism_Screening_Data_Combined.csv: No exact single-feature rule over the whole file; best is 'ASD if sum(A1..A10) >= 7' accuracy 0.933827 (majority baseline 0.703045); PARTIAL LEAKAGE: within age bands the item-sum rule is exact for 5021 of 6075 rows / Toddler Autism dataset July 2018.csv: LEAKAGE: label reproduced exactly by 'ASD if sum(A1..A10) >= 4' (accuracy 1.0) / Toddler Autism dataset July 2018.csv is embedded (rows 5021-6074) in Autism_Screening_Data_Combined.csv with 728 labels changed	tabular baselines only (LR, RF, GB, MLP) - NOT the CNN/ViT hybrids
D06_early_autism	early_autism	questionnaire / screening table	asd.csv 1000x11	no subject IDs beyond per-row ID columns (asd.csv: Child_ID); cannot tell whether rows	asd.csv: {'No': 589, 'Yes': 411}	asd.csv: 0 full / 0 excl. ID	asd.csv: No exact single-feature rule over the whole file; best is 'ASD if Age >= 35' accuracy 0.601 (majority baseline	tabular baselines only (LR, RF, GB, MLP) - NOT the CNN/ViT hybrids

id	folder name	modality	n samples	n subjects	classes and counts	duplicates found	leakage findings	suitable for
				are distinct people			0.589)	

5. Problems found

1. **D01 - no child ID on 901,786 of 2,252,943 gaze rows** {'Unidentified(Neg)': 878362, 'Unidentified(Pos)': 23424} - `D01_eye_tracking_audit.json`
2. **D01 - duplicated recordings across files:** 10 file pairs share rows, e.g. 8.csv/11.csv: 114,285; 10.csv/25.csv: 51,828; 1.csv/16.csv: 34,630. Identified rows 1,351,157 -> 1,103,660 after dedupe (247,497 removed, 18,490 of them within a single file) - `D01_eye_tracking_audit.json`, `D01_dedupe_key_check.json`
3. **D01 - dedupe key must be numeric:** a string key removes only 205,275 identified duplicate rows vs 247,497 with numeric parsing - `D01_dedupe_key_check.json`
4. **D01 - participants without gaze data:** [12, 16] (both ASD), leaving 57 children {'ASD': 27, 'TD': 30} - `D01_eye_tracking_audit.json`
5. **D01 - CARS Score encodes the label:** missing by class {'ASD': {'cars_missing': 0, 'n': 29}, 'TD': {'cars_missing': 30, 'n': 30}} - `D01_eye_tracking_audit.json`
6. **D01 - 9 different column layouts across 25 files** (37-58 columns); sampling interval varies by file (median 16.7 ms overall, p95 44.0 ms) - `D01_eye_tracking_audit.json`
7. **D01 - small sample:** 57 children -> 5-6 per test fold (2-3 ASD); fold metrics will have high variance - `D01_cv_feasibility.json`
8. **D03 is a copy of D02:** 2,541 byte-identical + 399 re-saved copies (nearest D02 split {'valid': 99, 'test': 300}); all 2,940 D03 images match D02 with the same class - `D02_facial_images_A_audit.json`
9. **D02 - near-duplicates across the original splits:** 11 pHash<=4 pairs, 5 across splits {'train|valid': 2, 'test|train': 3}; exact md5 groups 0 - `D02_facial_images_A_audit.json`
10. **D02/D03 - no subject IDs** (filenames `<class>.<n>.jpg` / `<n>.jpg`, no metadata): the same child may occur in several images; results can only be image-level - `D02_facial_images_A_audit.json`
11. **D02 - inconsistent class folder name** ['Autistic', 'Non-Autistic'] in test vs ['Autistic', 'Non_Autistic'] in train; 10 result-figure PNGs in the folder root (not data) - `D02_facial_images_A_audit.json`
12. **D02 - image size varies widely:** width 114-2471 px, height 126-2872 px, 2,557 distinct sizes; unreadable files: 0 - `D02_facial_images_A_audit.json`
13. **D04 - test.csv has no label** (shape [200, 21], label columns []) - `D04_autism_prediction_audit.json`
14. **D04 - class imbalance:** train {'0': 639, '1': 161} (20.1% positive); '?' placeholders {'ethnicity': 203, 'relation': 40} - `D04_autism_prediction_audit.json`
15. **D05 - label leakage (Toddler):** 'ASD if sum(A1..A10) >= 4' accuracy 1.0, confusion [[326, 0], [0, 728]]; item sum == Qchat-10-Score on 1054/1054 rows - `D05_toddler_screening_audit.json`
16. **D05 - wrong labels in Combined file:** Toddler rows embedded at rows 5021-6074, all labelled {'no': 1054} vs original {'no': 326, 'yes': 728} -> 728 conflicts - `D05_toddler_screening_audit.json`
17. **D05 - partial leakage in Combined file:** No exact single-feature rule over the whole file; best is 'ASD if sum(A1..A10) >= 7' accuracy 0.933827 (majority baseline 0.703045); PARTIAL LEAKAGE: within age bands the item-sum rule is exact for 5021 of 6075 rows - `D05_toddler_screening_audit.json`
18. **D05 - duplicates:** Combined 847 fully duplicated rows; Toddler 79 duplicates when Case_No is ignored - `D05_toddler_screening_audit.json`

19. **D05 - class imbalance (Toddler):** {'Yes': 728, 'No': 326} (69.1% ASD) - `D05_toddler_screening_audit.json`
20. **D06 - empty column** ['Unnamed: 10']; weak signal: best single rule 'ASD if Age >= 35' 0.601 vs majority baseline 0.589 - `D06_early_autism_audit.json`
21. **Licences:** no licence file in any of the 6 folders; only D01 `Citation.txt` and the D05 description (.docx) exist - `data_registry/*_licence_citation.txt`
22. **No unreadable images, no zip/rar archives** in any folder - `*_audit.json` (`n_unreadable`, `archives`)

6. Dataset decisions

Full text with evidence: data_registry/DATASET_DECISIONS.md. No CONTRADICTION found.

- DD-01 - D01 eye-tracking is the PRIMARY dataset for the 10 hybrid architectures
- DD-02 - D01 duplicated rows across files are removed in Stage 2 BEFORE any clip or window is created
- DD-03 - D01 CARS Score is never a feature, filter or stratification variable
- DD-04 - D01 10-fold CV = StratifiedGroupKFold(n_splits=10, shuffle=True, random_state=42)
- DD-05 - D02 facial images (DFDK folder) is the SECONDARY dataset for the 10 hybrid architectures
- DD-06 - D03 AutismDataset is EXCLUDED from training: it is a copy of D02
- DD-07 - D02 near-duplicates are grouped, and CV uses StratifiedGroupKFold on those groups
- DD-08 - D04 autism prediction: only the 800 labelled train rows are used
- DD-09 - D05 toddler screening: the label is defined by the A1–A10 score
- DD-10 - D06 early autism: tabular baselines only, 10-fold StratifiedKFold; the empty column is dropped
- DD-11 - No datasets are merged, and no predictions are combined across children or datasets
- DD-12 - Shortcut control (in Stage 8): tracking quality as a baseline for D01

7. Tests

Command: `.venv/bin/python -m pytest tests/test_stage01.py -v` (output: `logs/stage01_pytest.txt`)

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/score/asd_hybrid_10fold/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/score/asd_hybrid_10fold
plugins: anyio-4.15.1
collecting ... collected 24 items

tests/test_stage01.py::test_skeleton_dir_exists[configs] PASSED          [  4%]
tests/test_stage01.py::test_skeleton_dir_exists[src/data] PASSED         [  8%]
tests/test_stage01.py::test_skeleton_dir_exists[src/models] PASSED       [ 12%]
tests/test_stage01.py::test_skeleton_dir_exists[src/training] PASSED     [ 16%]
```

```
tests/test_stage01.py::test_skeleton_dir_exists[src/evaluation] PASSED   [ 20%]
tests/test_stage01.py::test_skeleton_dir_exists[src/utils] PASSED        [ 25%]
tests/test_stage01.py::test_skeleton_dir_exists[scripts] PASSED          [ 29%]
tests/test_stage01.py::test_skeleton_dir_exists[tests] PASSED            [ 33%]
tests/test_stage01.py::test_skeleton_dir_exists[data_registry] PASSED    [ 37%]
tests/test_stage01.py::test_skeleton_dir_exists[splits] PASSED           [ 41%]
tests/test_stage01.py::test_skeleton_dir_exists[cache] PASSED            [ 45%]
tests/test_stage01.py::test_skeleton_dir_exists[results] PASSED          [ 50%]
tests/test_stage01.py::test_skeleton_dir_exists[reports] PASSED          [ 54%]
tests/test_stage01.py::test_skeleton_dir_exists[logs] PASSED             [ 58%]
tests/test_stage01.py::test_results_readme_and_gitignore PASSED          [ 62%]
tests/test_stage01.py::test_audit_json_loads_and_nonzero[D01_eye_tracking] PASSED [ 66%]
tests/test_stage01.py::test_audit_json_loads_and_nonzero[D02_facial_images_A] PASSED [ 70%]
tests/test_stage01.py::test_audit_json_loads_and_nonzero[D03_facial_images_B] PASSED [ 75%]
tests/test_stage01.py::test_audit_json_loads_and_nonzero[D04_autism_prediction] PASSED [ 79%]
tests/test_stage01.py::test_audit_json_loads_and_nonzero[D05_toddler_screening] PASSED [ 83%]
tests/test_stage01.py::test_audit_json_loads_and_nonzero[D06_early_autism] PASSED [ 87%]
tests/test_stage01.py::test_registry_exists PASSED                       [ 91%]
tests/test_stage01.py::test_raw_data_unmodified PASSED                   [ 95%]
tests/test_stage01.py::test_cuda_available PASSED                        [100%]

============================== 24 passed in 1.06s ==============================
```

Raw-data snapshot comparison (`logs/stage01_snapshot_compare.txt`):

```
$ cmp data_registry/raw_snapshot_start.tsv data_registry/raw_snapshot_end.tsv
IDENTICAL: 5944 entries (path, kind, size, mtime_ns), sha256 start=51873a4ec225bc64 end=51873a4ec225bc64
```

8. Open items for Stage 2 (data preparation + 10-fold split manifests + leakage tests)

1. D01: drop Unidentified rows, exclude IDs [12, 16], dedupe with the numeric DD-02 key, and report rows before/after (expected identified rows 1,351,157 -> 1,103,660); harmonise the 9 column layouts; define how gaze-image sequences (clips/windows) are rendered from gaze rows.
2. D01: save the StratifiedGroupKFold(10, shuffle, seed 42) participant manifest in `splits/`; assert each test fold has >=1 ASD and >=1 TD and that no participant appears in two folds.
3. D02: pool the 2,940 images, normalise class names, build pHash<=4 groups, save the StratifiedGroupKFold manifest; assert no md5 or near-duplicate group crosses folds.
4. D03: excluded; add a test that no D03 path ever appears in a manifest.
5. D04 / D05-Toddler / D06: save StratifiedKFold(10, shuffle, seed 42) manifests; D05 manifests for experiments (a) all features and (b) without A1-A10/Qchat; treat '?' as missing in D04; drop `Unnamed: 10` in D06.
6. Leakage tests (pytest): no ID/group overlap between train and test in any fold; CARS never in any D01 feature list; D05 experiment (b) feature list contains no A1-A10/score; DD-11 (no cross-dataset merges).

7. Stage 8 shortcut control (DD-12): compute the per-child tracking-quality ratio in Stage 2 so it is available later.

Git: git is not installed (`git --version` -> command not found)

STAGE 1 DECISION: FAIL - git commits not made: git is not installed (`git --version` -> command not found)

Dataset decisions (Stage 1)

Every number below was loaded from the audit outputs, never typed from memory. The printout of these values is in `logs/stage01_facts.txt`. Sources: `data_registry/<id>_audit.json` (from `scripts/01_audit_datasets.py`), `data_registry/D01_dedupe_key_check.json` (`scripts/01_check_dd02_key.py`), `data_registry/D01_cv_feasibility.json` (`scripts/01_check_cv_feasibility.py`). Raw data root: `/home/score/Documents/AI_project/dataset`.

No CONTRADICTION was found. All 12 decisions are kept as specified. DD-02 adds an implementation requirement that the evidence calls for, and does not change the decision.

DD-01 - D01 eye-tracking is the PRIMARY dataset for the 10 hybrid architectures

- Unit of analysis is the **participant (child)**.
- Rows whose participant is “Unidentified” are dropped.
- Participants without gaze rows are excluded.

Evidence (`D01_eye_tracking_audit.json`): - `total_gaze_rows` = 2,252,943 in 25 CSV files. - `non_numeric_participant_values_rows` = {Unidentified(Neg): 878,362; Unidentified(Pos): 23,424}, a total of 901,786 rows (40.0%) with no child ID. These rows cannot be assigned to a child or a label, so they are dropped. - `metadata_n_participants` = 59 ({TD: 30, ASD: 29}). - `participants_with_metadata_and_gaze` = 57 ({ASD: 27, TD: 30}). - `participants_metadata_but_no_gaze` = **[12, 16]**, both ASD. These two are excluded. - `participant_ids_gaze_but_no_metadata` = []. - The analysis set is therefore **57 children (27 ASD, 30 TD)**.

DD-02 - D01 duplicated rows across files are removed in Stage 2 BEFORE any clip or window is created

- Dedupe key = participant + recording timestamp + trial + gaze coordinates.
- Rows before and after dedupe must be reported.

Evidence: - `D01_eye_tracking_audit.json` → `duplicate_check_dd02_key.file_pairs_sharing_keys` lists 10 file pairs that share rows. The largest:

file pair	shared keys	rows in each file
`8.csv` / `11.csv`	114,285	114,285 and 114,285 (the same content, despite different column layouts: 39 vs 52 columns)
`10.csv` / `25.csv`	51,828	51,829 and 119,064
`1.csv` / `16.csv`	34,630	34,630 and 110,967 (file 1 is entirely inside file 16)
`21.csv` / `22.csv`	24,266	
`21.csv` / `24.csv`	19,822	
`14.csv` / `21.csv`	10,469	
`4.csv` / `21.csv`	9,728	
`5.csv` / `21.csv`	104	
`9.csv` / `21.csv`	93	
`18.csv` / `21.csv`	93	

- `file_md5_identical_groups` = []. No two files are byte-identical; the duplication is in their content.
- `D01_dedupe_key_check.json`, identified rows only, using the key with numeric parsing: **1,351,157 rows before → 1,103,660 after** (247,497 removed, 18,490 of them duplicates within the same file).

Implementation requirement (from the evidence): Stage 2 must compare the key **numerically**, not as text: RecordingTime rounded to 3 decimal places, Point of Regard X/Y rounded to 4, and “-” treated as missing. Comparing the raw strings removes only 205,275 identified rows (1,145,882 remain), which misses 42,222 true duplicates.

DD-03 - D01 CARS Score is never a feature, filter or stratification variable

- From the metadata, only ParticipantID and Class are used.
- Age and Gender are kept for reporting only.

Evidence: `D01_eye_tracking_audit.json` → `cars_missing_by_class` = {ASD: 0 missing of 29; TD: 30 missing of 30}. Whether CARS is present identifies the class perfectly, so it encodes the label.

DD-04 - D01 10-fold CV = StratifiedGroupKFold(n_splits=10, shuffle=True, random_state=42)

- Folds are grouped by participant and stratified by Class.
- Every test fold must contain at least one ASD child and one TD child.

Evidence: `D01_cv_feasibility.json` (a feasibility check only; no manifest was saved). - Over the 57 children with gaze rows, every test fold has 5–6 children: 2–3 ASD and 3 TD. - `every_fold_has_asd_and_td` = true. - Stage 2 must re-assert this on the saved manifest.

DD-05 - D02 facial images (DFDK folder) is the SECONDARY dataset for the 10 hybrid architectures

- The original train/valid/test folders are pooled (2,940 images) and re-split by 10-fold CV.

Evidence: `D02_facial_images_A_audit.json` - `count_per_split_x_class`: train 1,193/1,193, valid 127/127, test 150/150, which gives 2,940 images (1,470 autistic, 1,470 non-autistic). - `n_unreadable` = 0.

Two things Stage 2 must handle: - The test class folder is named `Non-Autistic` while train and valid use `Non_Autistic` (`class_folder_names_per_split(raw)`). Both normalise to `nonautistic`. - The 10 PNG files in the folder root (`CF.png`, `roc.png`, ...) are result figures from the original paper and are not samples.

Full folder name: `Autism Spectrum Disorder Detection using Facial Traits Dual Framework DenseRes with K-Fold Cross Validation (DFDK) for Enhanced Classification Accuracy.`

DD-06 - D03 AutismDataset is EXCLUDED from training: it is a copy of D02

Evidence: `D02_facial_images_A_audit.json` → `cross_dataset_duplicates` - `unique_md5_shared` = **2,541** byte-identical images. - **399** D03 images are re-saved copies (not byte-identical): - nearest pHash distance is 0 for 398 of them and 2 for 1; - all 399 have the same pixel size as their D02 match; - their nearest D02 matches are 300 test images and 99 valid images. - All 2,940 of 2,940 D03 images have a pHash ≤ 4 match in D02, and all 2,940 have the same class.

Using both datasets would put the same face in train and test.

DD-07 - D02 near-duplicates are grouped, and CV uses StratifiedGroupKFold on those groups

- Near-duplicate images (pHash Hamming distance ≤ 4) are merged into one group (connected components).
- Near-duplicates therefore never cross folds.
- D02 has **no subject IDs**, so the same child may appear in different photos across folds. This is a documented limitation: results are image-level, not child-level.

Evidence: `D02_facial_images_A_audit.json` → `near_duplicates_phash` - 11 pairs, forming 11 connected groups of 2. - 6 pairs fall within one original split and 5 cross splits ({train/valid: 2, test/train: 3}). - 0 pairs cross classes. - `exact_duplicates_md5.n_groups` = 0. - This gives 2,940 − 11 = 2,929 CV groups. - `subject_ids.finding`: the filenames are only `<class>.<index>.jpg` or `<index>.jpg`, and the folder has no metadata files.

DD-08 - D04 autism prediction: only the 800 labelled train rows are used

- test.csv (no labels) is not used.
- Tabular baselines only (LR, RF, GB, MLP), with 10-fold StratifiedKFold. D04 is not used for the CNN/ViT hybrids.

Evidence: `D04_autism_prediction_audit.json` - `train.csv` shape [800, 22], label `Class/ASD` = {0: 639, 1: 161}, so 20.1% positive. - `test.csv` shape [200, 21], `label_columns` = [], so it is unlabelled. - No exact leakage: the best single rule is `sum(A1_Score..A10_Score) >= 10`, with accuracy 0.8625 against a majority baseline of 0.79875. - `'?'` placeholders appear in train (ethnicity 203, relation 40). These are missing values and must be treated as such.

DD-09 - D05 toddler screening: the label is defined by the A1–A10 score

Any model that sees A1–A10 is reproducing a formula, not predicting autism. Two tabular experiments: - **(a)** all features, reported explicitly as “label-definitional / upper bound”; - **(b)** demographic and history features only, with A1–A10 and `Qchat-10-Score` removed. That leaves `Age_Mons, Sex, Ethnicity, Jaundice, Family_mem_with_ASD, Who completed the test`; `Case_No` is a row ID and is dropped.

The Combined file is EXCLUDED. Tabular baselines only.

Evidence (`D05_toddler_screening_audit.json`): - `Toddler Autism dataset July 2018.csv` (1,054 rows; Yes 728 / No 326): - The rule **`ASD if sum(A1..A10) >= 4`** has accuracy **1.0**, confusion [[326, 0], [0, 728]]. - `item_sum_equals_Qchat-10-Score`: 1,054 of 1,054 rows equal. - The dataset’s own description (`D05_toddler_screening_licence_citation.txt`, copied from `Toddler data description.docx`) says (verbatim, including its typo): “the class varaible was assigned automatically based on the score obtained by the user”. - `Autism_Screening_Data_Combined.csv` (6,075 rows) is excluded: - `cross_file_embedded_blocks`: all 1,054 Toddler rows appear, in order, at rows **5021–6074**, compared on A1–A10 and Sex. They carry the label `NO` for **1,054 of 1,054** rows, while the original labels are 326 No / 728 Yes, so **728 labels conflict**. - The file also has 847 fully duplicated rows. - Within the other age bands the label is exactly `item_sum >= 7`: accuracy 1.0 on 5,021 of 6,075 rows. - The Toddler file has 79 rows that are duplicates once Case_No is ignored. Stage 2 must report them.

DD-10 - D06 early autism: tabular baselines only, 10-fold StratifiedKFold; the empty column is dropped

Evidence: `D06_early_autism_audit.json` - `asd.csv` shape [1000, 11], `Diagnosed_ASD` = {No: 589, Yes: 411}. - `fully_empty_columns` = **[`Unnamed: 10`]**, missing in 1,000 of 1,000 rows. - `Child_ID` is unique per row and is not a feature. - No leakage: the best single rule is `Age >= 35`, with accuracy **0.601**, against a majority baseline of **0.589**.

DD-11 - No datasets are merged, and no predictions are combined across children or datasets

Each dataset is trained and reported separately.

Evidence: D02 and D03 are the same images (DD-06), and the D05 Combined file is a relabelled merge (DD-09). Both show how merging creates leakage and label conflicts. The datasets also do not share subjects or ID schemes: - D01 uses ParticipantID 1–59. - D02 has no IDs. - D04, D05 and D06 have only per-row IDs.

DD-12 - Shortcut control (in Stage 8): tracking quality as a baseline for D01

- For D01, the per-child tracking-quality ratio (fraction of invalid or lost samples) is evaluated alone as a baseline classifier.
- Every hybrid model must be compared against it.

Evidence: `D01_eye_tracking_audit.json` - `tracking_ratio_column_in_files` = 25 of 25 (`Tracking Ratio [%]` is present in every gaze file), and `Category Right` (e.g. Fixation, Blink, Separator) is present in all 9 column layouts. - Sampling is a median of 16.7 ms between samples (≈59.9 Hz), with p95 at 44.0 ms, so dropped samples are measurable. - Whether tracking quality differs by class is **UNKNOWN** at Stage 1. Measuring it is the purpose of DD-12.